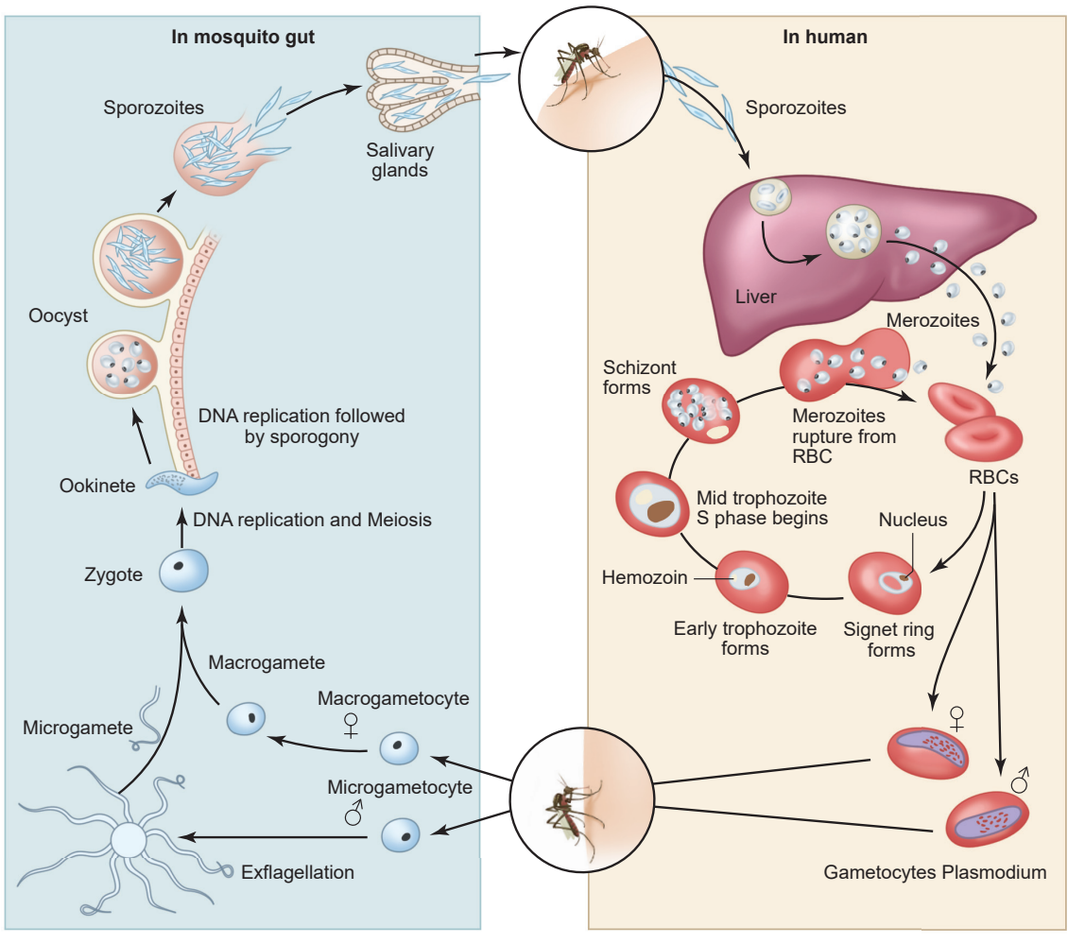


Fig. 7.3 Life cycle of *Plasmodium*

The early symptoms of malaria are headache, nausea and muscular pain.

The classic symptoms first develop with the synchronized release of merozoites, **haemozoin** toxin and erythrocyte debris into the blood stream resulting in malarial paroxysms – shivering chills, high fever followed by sweating. Fever and chills are caused partly by malarial toxins that induce macrophages to release tumour necrosis factor (TNF-α) and interleukin.

Prevention

It is possible to break the transmission cycle by killing the insect vector. Mosquitoes lay their eggs in water. Larvae hatch and develop in water but breathe air by moving to the surface. Oil can be sprayed over the water surface, to make it impossible for mosquito larvae and pupae to breathe.

Ponds, drainage ditches and other permanent bodies of water can be stocked

Table 7.3 Types of malaria

Sl. No.	Types of Malaria	Causative agent	Duration of Erythrocytic cycle
1	Tertian, benign tertian or vivax malaria	*P. vivax*	48 hours
2	Quartan malaria	*P. malariae*	72 hours
3	Mild tertian malaria	*P. ovale*	48 hours
4	Malignant tertian or quotidian malaria	*P. falciparum*	36 – 48 hours

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